Infrastructure-Free AR Headset Tracking System Using Thermal Sensors

CHANDAFA, Abraham Ernest

December 1, 2025

1 Introduction

Augmented Reality (AR) technology is among the most rapidly expanding fields of this century [1]. With vast applications in many fields, such as creating immersive 3D virtual tours for real estate, surgical simulations in medicine, and 3D gaming experiences [2], augmented reality creates an immersive experience by enabling real-time superimposition of digital information and virtual objects onto the physical world.

For a great immersive experience, one of the key tasks for Augmented Reality (AR) systems is to accurately determine the 6-Degree-of-Freedom (6-DoF) pose of the camera. As pointed out by Xu et al. [3], modern camera pose estimation techniques that are image-based can be categorised into two paradigms: **structure feature-based localisation**, which recovers the camera pose by establishing correspondences between features in the given query image and a 3D model of the scene, and **regression-based localisation**, where deep learning models, such as PoseNet, are trained to directly regress the camera pose from an input image in an end-to-end fashion.

To estimate camera pose, these traditional pose estimation techniques rely on tracking visible texture and patterns in the environment, a method that faces serious drawbacks in poorly textured environments [3]. As a solution, artificial physical patterns, such as QR codes or remotely projected patterns can be introduced to the environment in the absence of good texture and patterns [4]. But introducing physical patterns is often destructive and not ideal for most scenarios, whereas projecting patterns is inefficient for dynamic vision tasks because the projector moves with the camera, as illustrated in figure 1. As will be shown, a thermal camera and laser projector can achieve what a normal camera and projector cannot: draw a pattern that ”sticks” to the surface as the projector moves, and track that pattern in the thermal domain, achieving an effect similar to having transient but fixed physical patterns.

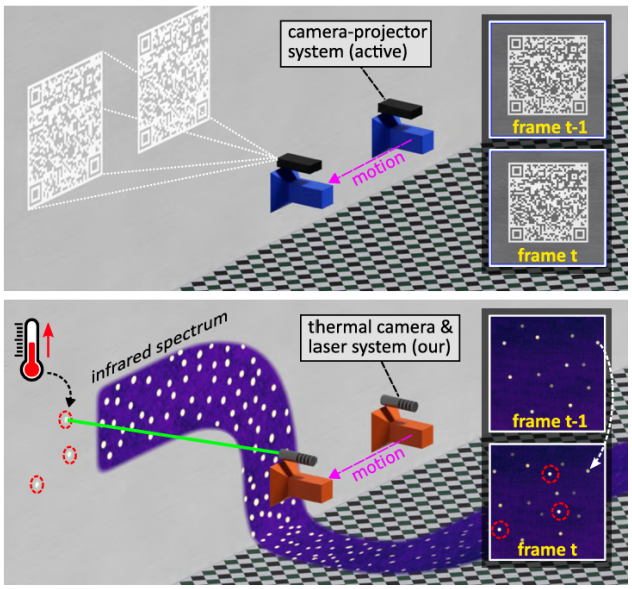

Figure 1: Projecting trackable thermal patterns. Adapted from [5]. (Top) The projected features cannot be used for reference between frames because the projector moves with the camera. (Bottom) The laser can heat up points, temporarily creating a heat pattern that 'sticks' on surfaces and can be tracked in the thermal domain.

1.1 Background

Thermal sensing and imaging have undergone a significant expansion in recent years, driven by their increasing adoption across diverse domains. As highlighted by Wilson et al. [6], machine learning has enabled the advancement of thermal sensing technology beyond traditional military and defence applications, where thermal cameras aid in identifying intruders in the dark, as well as in medical applications, where thermal imaging facilitates medical diagnosis.

One such pivotal advancement was introduced by Sheinin et al. [5], who proposed a novel framework for projecting trackable thermal patterns using a co-located thermal camera and steerable laser system. Their approach utilises the transient yet persistent nature of laser-induced thermal signature to "paint" features onto otherwise textureless surfaces. These heat patterns can be tracked by thermal cameras, overcoming a fundamental limitation of traditional projected patterns that move with the projector.

A fundamental challenge with thermal patterns, however, is heat diffusion; heat quickly evaporates and dissipates into the material, deforming the pattern. For a thermal pattern to be trackable, it must have a structure in space with trackable feature that persist for at least two consecutive video frames [5]. Sheinin has addressed this heat diffusion problem through a learning-based approach called **Projection-Diffusion Reversal** (PDR), where they trained a neural network to remove the newly added thermal points and reinforce existing points that have diffused, producing a consistent thermal pattern across frames.

While Sheinin et al. demonstrated the feasibility of thermal pattern projection for general computer vision tasks, its application to the specific, demanding problem of AR headset pose estimation remains unexplored.

1.2 Project Motivation

Motivated by the critical need for robust, infrastructure-free tracking for augmented reality headsets, this project builds on the foundational work of Sheinin et al. by adapting active thermal pattern projection for real-time estimation and tracking of the 6-degree-of-freedom (6-DoF) pose of an augmented reality (AR) headset.

As comprehensively surveyed by Zhou et al. [7] in their review of the ten years of International Symposium on Mixed and Augmented Reality (ISMAR) research, tracking has consistently been the most challenging and actively searched area in AR, constituting over 20% of all publications. See table 1 for reference. Traditional tracking methods still face fundamental limitations, despite decades of development:

- **Marker-based tracking**, while robust, requires frequent maintenance, has a limited range, and suffers from intermittent errors since it only provides location information when markers are in sight, rendering them impractical for dynamic, large-scale environments [7].
- **Natural feature tracking** methods struggle in texture-poor spaces where distinguishable objects that can be used as markers may not be present, which may lead to failure in tracking [3, 7].
- **Model-based approaches** require a burdensome modeling process, especially when used for creatiing detailed models for large, cluttered environments [7], thus limiting their scalability.

AR headsets are particularly sensitive to these limitations, since they must operate reliably in real-world environments, including dimly lit rooms, featureless corridors and hallways, and other visually challenging conditions.

By adapting Sheinin's thermal projection framework specifically for AR headset pose estimation, this work aims to overcome the critical limitations identified in the ISMAR survey and enable AR systems to work reliably under varying lighting conditions and on textureless surfaces.

1.3 Project Objectives

The goal of this project is to develop a prototype system demonstrating real-time 6-DoF pose estimation for an AR headset using actively projected thermal patterns. To achieve this, the following objectives have been defined:

Topics	% Papers	% Citations
Tracking	20.1	32.1
Interaction	14.7	12.5
Calibration	14.1	12.5
AR App.	144	12.5
Display	11.8	5.4
Evaluations	5.8	1.8
Mobile AR	6.1	7.1
Authoring	3.8	8.9
Visualization	4.8	5.4
Multimodal	2.6	0.0
Rendering	1.9	1.8

Table 1: Proportion of Highly Cited Papers. Adapted from [7]. This table considers all IS MAR/ISAR/ISMR/IWAR papers with an average citation rate of 5.0 or more per year.

1. **System Design and Integration:** Design and assemble a hardware prototype comprising a thermal camera, a steerable laser projector, and computing hardware, configured to be head-mountable. The system will adopt Sheinin's coaxial configuration, where "each camera pixel can be mapped to a corresponding outgoing projector ray" [5], ensuring predictable pattern projection regardless of scene geometry.

2. **Pattern Projection and Tracking:** Implement software to control the laser projector, creating sparse, trackable thermal dot patterns on surfaces in the environment. Following Sheinin's findings, discrete patterns will be used with "only a few dots per frame as opposed to 'continuous' patterns" [5] in order to maintain trackability and develop a tracker to follow these dots in the thermal video stream.

3. **Pose Estimation Pipeline:** Develop and implement a software pipeline that uses the tracked 2D positions of the thermal dots to compute the 6-DoF pose of the headset relative to the room. This extends Sheinin's structure-from-motion approach, which demonstrated that "the system continuously projects new 'heat points' on the object's surface during motion" to generate accurate camera trajectories [5].

4. **Performance Evaluation:** Quantitatively evaluate the system's performance in terms of tracking accuracy (translation and rotation error), latency, and robustness compared to traditional RGB-based tracking in both textured and textureless environments. The aim is to achieve performance comparable to Sheinin's indoor localisation experiment, which demonstrated "accurate tracks and loop closure" over a 20-meter trajectory [5].

1.4 Project Scope and Deliverables

This project will deliver a functional hardware prototype and software pipeline for infrastructure-free AR headset tracking using thermal sensors. The prototype will comprise a head-mounted rig featuring a thermal camera and laser projector in a coaxial configuration, capable of projecting sparse thermal dot patterns onto surfaces. The target environment is a controlled indoor space with texture-poor surfaces, and the system will perform real-time 6-DoF pose estimation. For deliverables, the project will deliver a calibrated hardware prototype as well as an open-source software repository on GitHub that contains an implementation of the complete tracking pipeline, including thermal dot projection, Projection-Diffusion Reversal network integration, feature tracking, and Visual SLAM. Finally, a comprehensive final project report that contains a quantitative performance evaluation, as well as a demonstration video that showcases the prototype in action, will be delivered.

1.5 Report Outline

The remainder of this report details the methodology in Section 2, explaining the steps involved while justifying the engineering choices made, followed by Section 3, which presents the progress of the project, including the status of implementation, as well as the difficulties encountered. Finally, Section 4 concludes by summarising key achievements, discussing encountered challenges and their mitigation, and outlining future work for system refinement and evaluation.

2 Methodology

This project adopts a simulation-first methodology by starting with a comprehensive software simulation of thermal pattern projection, diffusion, and tracking algorithms to minimise hardware dependencies and bottlenecks at the initial development phase. Physical hardware integration will commence after developing the core algorithms in simulation to ensure a robust algorithm validation and to reduce development risk. For software pipeline development, we will simulate thermal pattern projection while mimicking natural heat diffusion before looking into the Projection-Diffusion Reversal (PDR) network to reverse heat diffusion effects. After the software pipeline is developed, we will explore pose estimation and VSLAM integration.

2.1 Thermal Pattern Projection

The first step involves simulating sparse thermal dot patterns in software to match as closely as would appear in the physical environment. According to Sheinin, two conditions are necessary for

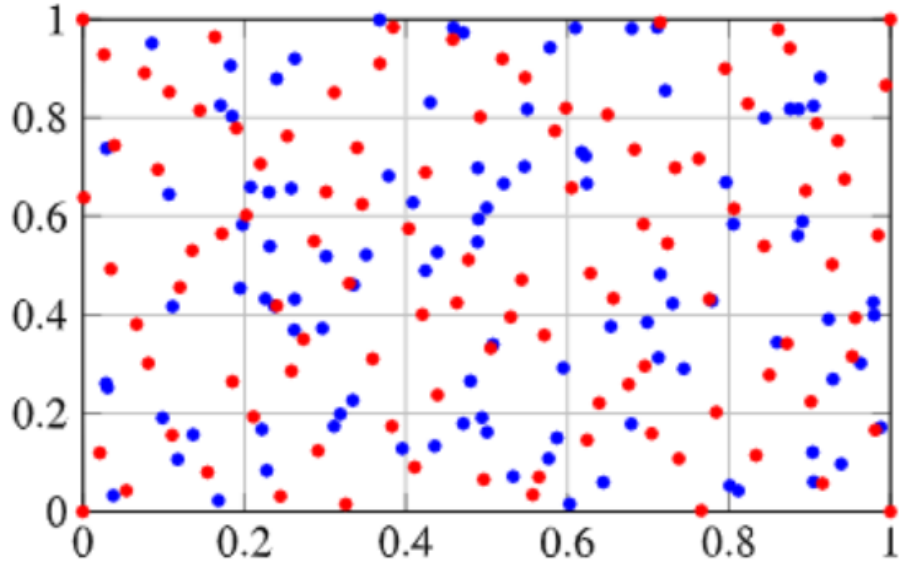

Figure 2: This figure shows the difference between a random number sequence (blue dots) and a Halton Sequence (red dots). Adapted from [9]. Halton sequences cover the space uniformly, while a random sequence produces clumps and uncovered regions.

a pattern to be trackable: the pattern must have a spacial structure that produces feature points that are trackable, and these trackable feature points must remain so for two or more consecutive video frames [5]. To simulate this, the Halton sequence will be used to generate the thermal dot patterns. The Halton sequence was chosen over a random sampling approach because it is a low-discrepancy sequence, or, in simple terms, a deterministic sequence of points that is designed to fill a given space as uniformly as possible, avoiding clumps produced by a purely random sampling [8]. Clumping may lead to large areas of the frame without trackable features. Figure 2 compares the random number sequence against the Halton sequence, further justifying the choice of the latter. Other than providing more uniformly distributed points, the Halton sequence has the advantage of being deterministic and computationally efficient.

We project a sparse pattern of K inclusively between 2 and 4 new points per frame, synchronised with typical a thermal camera's capture rate of 30 Hz. This sparse, discrete projection strategy is critical, as Sheinin et al. found that the signal-to-noise ratio (SNR) of individual thermal dots deteriorates rapidly, reducing trackability duration. [5]

To realistically simulate natural heat dissipation, each projected thermal dot's intensity evolves according to the heat diffusion equation. Following Sheinin et al., for a laser heating a surface, the temperature evolution follows:

$$\frac{1}{\alpha}\frac{\partial T}{\partial t} = \Delta T + (1-\rho)\frac{P_{laser}\delta}{k}e^{-\delta z} \tag{1}$$

where P_{laser} is the laser power, α is the material's thermal diffusivity, Δ is the Laplace operator, z is the depth into the material, ρ is the material's albedo at the laser wavelength, k and δ are the thermal conductivity and absorption coefficient, respectively. This equation captures both spatial diffusion (through ΔT) and the laser's energy deposition profile. For simulation purposes, we approximate this by applying progressively larger Gaussian blur kernels to previously projected points, with standard deviation $\sigma = \sqrt{2\alpha\Delta t}$, and exponentially decaying intensity based on material properties.

2.2 Dynamic Tracking with Projection Diffusion Traversal

In the previous section, we showed how simulated thermal dot patterns diffuse with time, adhering to natural physical phenomena. This is what causes a need to constantly replace the diffusing and evaporating patterns with new ones. Regardless, we still have to computationally reverse the effects of heat diffusion and remove inconsistent new points between frames, a process called projection-diffusion traversal (PDR) [5].

The PDR process starts with preprocessing the thermal frames through normalisation to take into account the varying ambient temperatures. We scale each raw thermal frame $I_{\text{raw}}(x, f)$ to the range $[0, 1]$ using the equation:

$$I(x, f) = \frac{I_{\text{raw}}(x, f) - I_{\text{min}}}{I_{\text{max}} - I_{\text{min}}} \tag{2}$$

where I_{min} and I_{max} are determined from scene statistics across multiple frames. This normalisation maintains the camera's approximately affine response to temperature variations around room temperature.

The PDR network will output a residual image $F_\theta(I(x, f + m) \| P_{f,m}, m)$ that, when subtracted from the input frame, produces the diffusion-reversed output:

$$I_{\text{pdr}}(x, f + m) = I(x, f + m) - F_\theta(I(x, f + m) \| P_{f,m}, m) \tag{3}$$

This operation simultaneously removes newly projected points (which appear as positive values in the residual) while reinforcing diffused existing features (which appear as negative values).

For training, we collect sequences of static scenes with alternating periods of pattern projection and cooling. The loss function minimises the difference between the original frame $I(x, f)$ and the PDR output:

$$\mathcal{L} = \mathbb{E}_x \| I(x, f) - I_{\text{pdr}}(x, f + m) \|_2^2 \tag{4}$$

To generalise to dynamic scenes, random spatial shifts proportional to $\|t_n - t_{f+m}\|$ are applied

to the dot positions during training.

The output is integrated into the tracking pipeline in two complementary ways. First, the fully reversed frame $I_{\text{pdr}}(x, f+m)$ is used to track existing points through standard Lucas-Kanade optical flow. Second, the positive components of the residual image:

$$I_{\text{new}}(x, f+m) = \max(F_\theta(I(x, f+m)), 0) \tag{5}$$

are used to detect and initialise tracking for newly projected points, maintaining continuous feature tracks and overcoming the thermal patterns' transient nature [5].

2.3 Pose Estimation and VSLAM Integration

To transform the transient thermal patterns into robust 6-DoF pose estimates, our system will integrate three key components: thermal feature detection, inter-frame feature association, and VSLAM-based pose estimation.

For feature detection, we will use the Shi-Tomasi corner detector [10] in the thermal domain. Unlike SIFT or SURF detectors, Shi-Tomasi excels in tracking points across sequential frames, whereas SIFT and SURF are better suited for natural feature matching [11].

For feature association between consecutive frames, we implement the Kanade-Lucas-Tomasi (KLT) optical flow algorithm [12]. Compared to Kalman filtering, KLT is preferred for this application because it directly leverages spatial intensity gradients rather than motion models. For each detected Shi-Tomasi point p_i in frame $I(x, f)$, KLT computes its displacement vector Δp_i that minimizes the brightness error:

$$\Delta p_i = \arg\min_{\Delta p} \sum_{x \in N(p_i)} \|I(x + \Delta p, f+1) - I(x, f)\|^2 \tag{6}$$

where $N(p_i)$ is a local neighborhood around point p_i. The PDR-corrected thermal frames (Section 2.2) provide a consistent appearance across frames, significantly improving KLT tracking accuracy by maintaining the brightness constancy assumption.

The tracked features will feed into our VSLAM pipeline, which will extend ORB-SLAM3 [13] to handle thermal features. Whereas conventional VSLAM systems rely on persistent natural features, our implementation will involve complete monitoring of the thermal points' lifecycles. Each thermal point will be uniquely identified at projection time, and its expected lifetime will be modelled.

For 6-DoF pose estimation, we solve the Perspective-n-Point (PnP) problem using the EPnP algorithm [14] with RANSAC outlier rejection. Given $n \geq 4$ 2D-3D correspondences between tracked thermal points and their triangulated 3D positions, EPnP efficiently computes the rotation

matrix $\mathbf{R}$ and translation vector $\mathbf{t}$ that minimize reprojection error:

$$\min_{\mathbf{R},\mathbf{t}} \sum_{i=1}^{n} \|\pi(\mathbf{R}\mathbf{P}_i + \mathbf{t}) - \mathbf{p}_i\|^2 \tag{7}$$

where $\mathbf{P}_i$ are 3D points, $\mathbf{p}_i$ are their 2D projections, and $\pi(\cdot)$ is the camera projection function.

3 Progress Update

3.1 Current Progress

To date, significant progress has been made on the simulation-based implementation of thermal pattern projection, which forms the foundation of our infrastructure-free AR tracking system, although no results have been produced to date.

3.2 Difficulties Encountered

Several significant challenges have emerged during the implementation phase:

1. Material Property Variations: The most significant challenge has been accurately simulating heat diffusion across different materials. Our initial simulation assumed uniform material properties, but for real-world physical materials, characteristics vary greatly, and it is hard to identify the parameters.

2. Computational Complexity of Physical Simulation: Implementing the full heat diffusion equation (Equation 1) with accurate boundary conditions proved computationally challenging.

3. Thermal Camera Noise Modelling: Real thermal cameras show complex noise patterns (fixed pattern noise, temporal noise, etc.) that are difficult to simulate accurately.

4. Learning Curve: The vast content that needs to be learned and implemented requires significant time investment in literature review and skill development, making the overall progress slower than expected.

To mitigate these challenges, the following strategies were adopted:

- **Adaptive Parameter Framework:** Instead of a single fixed diffusion model, we are developing a parameter estimation module that can automatically adjust diffusion parameters based on observed thermal response in the first few frames.

- **Hybrid Simulation Approach:** We're implementing a hybrid simulation method that uses physics-based equations for the initial heat dissipation before switching to the computationally efficient Gaussian approximation for subsequent frames.

3.3 Remaining Work Plan and Schedule

Based on current progress, the project is slightly behind the initial timeline but remains on track for completion by the deadline.

4 Conclusion

The aim of this project is to develop an infrastructure-free AR headset tracking system using active thermal pattern projection. One critical limitation in current AR technology involves difficulty tracking in textureless environments. By adapting Sheinin et al.'s thermal projection framework [5] specifically for AR headset pose estimation, we seek to enable robust and infrastructure-free 6-DoF tracking. Next steps include completing the PDR implementation and feature tracking pipeline, integration with the VSLAM framework, and finally, hardware integration.

Accurately simulating real-world thermal behaviour has proven to be a significant challenge. However, the simulation-first approach is valuable in identifying and addressing complications before they arise in hardware deployment. The end goal is to deliver a functional prototype that demonstrates the viability of thermal pattern projection for infrastructure-free AR tracking by the project completion date.

The successful implementation of this system would signify a milestone in the field of AR tracking, potentially enabling new applications in environments where conventional tracking methods fail.

References

[1] T. A. Syed, M. S. Siddiqui, H. B. Abdullah, S. Jan, A. Namoun, A. Alzahrani, A. Nadeem, and A. B. Alkhodre, "In-depth review of augmented reality: Tracking technologies, development tools, ar displays, collaborative ar, and security concerns," *Sensors*, vol. 23, p. 146, dec 2023. Academic Editors: Calin Gheorghe Dan Neamtu, Radu Comes, Jing-Jing Fang and Dorin-Mircea Popovici.

[2] M. S. Siddiqui, T. A. Syed, A. Nadeem, W. Nawaz, and A. Alkhodre, "Virtual tourism and digital heritage: an analysis of vr/ar technologies and applications," *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 7, 2022.

[3] M. Xu, Y. Wang, B. Xu, J. Zhang, J. Ren, Z. Huang, S. Poslad, and P. Xu, "A critical analysis of image-based camera pose estimation techniques," *Neurocomputing*, vol. 570, p. 127125, 2024.

[4] J. Salvi, S. Fernandez, T. Pribanic, and X. Llado, "A state of the art in structured light patterns for surface profilometry," *Pattern Recognition*, vol. 43, no. 8, pp. 2666–2680, 2010.

[5] M. Sheinin, A. Sankaranarayanan, and S. G. Narasimhan, "Projecting trackable thermal patterns for dynamic computer vision," in *Proc. IEEE/CVF CVPR*, 2024.

[6] A. N. Wilson, K. A. Gupta, B. H. Koduru, A. Kumar, A. Jha, and L. R. Cenkeramaddi, "Recent advances in thermal imaging and its applications using machine learning: A review," *IEEE Sensors Journal*, vol. 23, no. 4, pp. 3395–3407, 2023.

[7] F. Zhou, H. B.-L. Duh, and M. Billinghurst, "Trends in augmented reality tracking, interaction and display: A review of ten years of ismar," in *2008 7th IEEE/ACM International Symposium on Mixed and Augmented Reality*, pp. 193–202, 2008.

[8] L. Kocis and W. J. Whiten, "Computational investigations of low-discrepancy sequences," *ACM Trans. Math. Softw.*, vol. 23, p. 266–294, June 1997.

[9] T. Keßler, C. Kunde, N. Mertens, D. Michaels, and A. Kienle, "Global optimization of distillation columns using surrogate models," *SN Applied Sciences*, vol. 1, 10 2018.

[10] J. Shi and C. Tomasi, "Good features to track," *Proceedings / CVPR, IEEE Computer Society Conference on Computer Vision and Pattern Recognition. IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, vol. 600, 03 2000.

[11] P. Tueller, R. Kastner, and R. Diamant, "A comparison of feature detectors for underwater sonar imagery," in *OCEANS 2018 MTS/IEEE Charleston*, pp. 1–6, 2018.

[12] J.-Y. Bouguet, "Pyramidal implementation of the lucas kanade feature tracker: Description of the algorithm," tech. rep., Intel Corporation, Microprocessor Research Labs, 2000. Available online: `https://docs.google.com/viewer?url=http://robots.stanford.edu/cs223b04/algo_tracking.pdf`.

[13] C. Campos, R. Elvira, J. J. G. Rodriguez, J. M. M. Montiel, and J. D. Tardos, "Orb-slam3: An accurate open-source library for visual, visual–inertial, and multimap slam," *IEEE Transactions on Robotics*, vol. 37, p. 1874–1890, Dec. 2021.

[14] V. Lepetit, F. Moreno-Noguer, and P. Fua, "Epnp: An accurate o(n) solution to the pnp problem," *International Journal of Computer Vision*, vol. 81, 02 2009.